

SHIVAM LALAKIYA

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Professional Summary

- Impact-driven Software Engineer and Data Scientist with over 5 years of experience architecting end-to-end data pipelines, developing applications for clinical research, and automating enterprise-scale workflows. Proficient in Python, Java, and shell scripting, with expertise in medical imaging informatics, data modeling, and orchestration (Apache Airflow, Docker, CI/CD). Active contributor to prominent open-source projects (including **LlamaIndex** and **PyMC-Marketing**), with experience in community collaboration and upstream code reviews. Proven track record of designing robust software solutions, managing release testing, and delivering highly reliable computational pipelines that directly optimize research operations and clinical workflows.

Experience

Washington University in St. Louis

Feb 2024 – Present

Data Analyst / Software Engineer, Grateful Patient Program

St Louis, MO

- Architected computational pipelines and programmatic suites (**Python, ARIMA, LSTM**) to model enterprise data, ensuring sound implementation of software architecture for high-stakes institutional strategy.
- Engineered a scalable, **object-oriented ETL automation framework**, designing data models and migrations that reduced a 4-hour manual process to under 5 minutes while maintaining strict data integrity.
- Orchestrated robust release workflows and data quality pipelines using **Apache Airflow**, writing comprehensive test plans to monitor data integrity and drive a **20% decrease** in downstream defects.
- Architected a context-aware **AI chatbot** interfacing directly with enterprise databases to automate data retrieval; currently scaling the solution across University Advancement (UA), reducing data request turnaround times from **days to minutes**.
- Collaborated with cross-functional technical teams to conduct gap analysis on legacy systems, designing holistic scoring algorithms that integrated seamlessly with existing enterprise data infrastructure.

Invicro, LLC.

Sept 2023 – Feb 2024

Associate Data Manager / Software Developer

Boston, MA

- Designed and implemented core functionalities for **medical imaging informatics** workflows, engineering automated pipelines for pathological image transfer and data management infrastructure.
- Managed end-to-end processing and deployment of **Computer Vision models** for biomarker identification, navigating complex computational pipelines for clinical research data.
- Architected data transformation schemas using **DBT (Data Build Tool)** and **SQL**, triaging defects and streamlining workflows to achieve a **25% reduction** in reporting errors for study sponsors.

Ring Therapeutics

May 2022 – Dec 2022

Data Science Co-op

Boston, MA

- Architected an automated ETL pipeline within a **Docker container** environment, applying robust orchestration principles to automate **90%** of data loading, migration, and formatting processes.
- Developed and fine-tuned machine learning models (**Python, NLP**) for gene prediction, utilizing advanced data structures and algorithms to generate vectors for targeted cell therapy (**0.75 F1-score**).
- Led the testing and integration of a **Protein-BERT model**, participating in code reviews and employing rigorous software testing methodologies to ensure accurate classification and novel motif identification.

IIT Madras

Jan 2020 – Jan 2021

Research Assistant

Chennai, India

- Applied core computer science concepts to develop a deep learning-based caching application, outperforming traditional algorithms (LRU, LFU) with a **130% improvement in system hit rates**.
- Engineered programmatic data processing solutions to clean and standardize over **100 GB** of time-series user data, achieving a **90% success rate** in system request forecasting.

Technical Skills

Languages: Java, Python, SQL, Shell Scripting, JavaScript, R, MATLAB

Software Engineering & Architecture: Object-Oriented Programming (OOP), Data Migrations, Data Modeling, RESTful APIs, CI/CD, Git, Release Management, Unit/Regression Testing

Cloud & Infrastructure: Linux/RHEL Administration, AWS, Docker, Apache Airflow, DBT, Apache Kafka

Domain Expertise: Medical Imaging Informatics, Computer Vision, Bioinformatics, Enterprise ETL Pipelines, Predictive Algorithms

Education

Northeastern University, Boston, MA <i>Master of Science in Data Analytics Engineering</i>	Sept 2021 – July 2023 <i>GPA: 3.8/4.0</i>
Sardar Vallabhbhai National Institute of Technology (NIT Surat), India <i>Bachelor of Technology in Electronics and Communication Engineering</i>	July 2017 – May 2021 <i>GPA: 8.1/10</i>

Publications

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- **Lalakiya, S.** (2025) *AI for Advancement: Predictive Donor Analytics and Fundraising Intelligence at Scale*
[doi:10.1109/ICCED68324.2025.11325064](https://doi.org/10.1109/ICCED68324.2025.11325064)
 - **Lalakiya, S.** (2025) *AI-Driven Observability for Real-Time Incident Management*
[doi:10.1109/ICRTEECT67512.2025.11448853](https://doi.org/10.1109/ICRTEECT67512.2025.11448853)
 - **Lalakiya, S.** (2025) *Enterprise AI Agents: Secure, Scalable, and Autonomous Intelligence for the Modern Workforce*
[doi:10.1109/ICCED68324.2025.11324763](https://doi.org/10.1109/ICCED68324.2025.11324763)

Extra-Curricular Activities

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- **Student Success Guide at Northeastern University:** Mentored **100+ students** per semester on academic, software engineering, and career strategies, contributing to an **85% rate** of students achieving career advancement.
 - **Math & Algorithm Tutor:** Tutored university-level technical courses including calculus, linear algebra, and probability to enhance student comprehension of core computational and mathematical concepts.